

HOYBUKTMOEN KIRKENES, Norway

WMO: 010890

Lat: 69.726N

Lon: 29.893E

Elev: 283

StdP: 14.55

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.2	-10.8	-20.6	1.8	-15.5	-14.8	2.5	-10.6	32.2	26.8	29.2	25.3	7.2	140	0.823

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	73.5	59.2	68.6	58.0	64.5	55.8	61.4	70.3	58.8	66.5	56.6	63.5	11.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.2	70.6	65.0	55.0	65.1	62.4	53.0	60.4	59.8	27.5	70.1	25.7	66.2	24.3	63.7	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 27.0	23.7	21.1	DB	-20.7	79.5	4.4	5.1	-23.9	83.2	-26.4	86.2	-28.9	89.1	-32.1	92.8	(4)
(5)			WB	-21.0	65.3	4.3	4.8	-24.1	68.7	-26.6	71.5	-29.0	74.2	-32.2	77.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.8	13.5	13.3	20.9	29.4	39.0	46.9	54.2	52.3	45.2	34.4	23.4	19.3
	DBStd	16.45	11.93	11.94	8.68	6.95	6.75	6.50	6.70	5.87	5.84	6.92	9.08	10.43
	HDD50	6632	1131	1028	901	618	349	136	28	38	168	485	798	952
	HDD65	11775	1596	1448	1366	1068	806	543	343	395	595	950	1248	1417
	CDD50	345	0	0	0	0	8	44	159	111	23	0	0	0
	CDD65	12	0	0	0	0	0	1	8	3	0	0	0	0
	CDH74	115	0	0	0	0	3	5	86	21	0	0	0	0
	CDH80	16	0	0	0	0	0	0	15	1	0	0	0	0
Wind	WSAvg	10.1	10.9	11.2	11.7	10.0	9.7	9.7	8.5	8.0	9.4	10.4	10.6	11.5
Precipitation	PrecAvg	18.4	1.3	1.3	0.9	0.9	1.1	2.0	2.4	2.3	1.7	1.8	1.3	1.3
	PrecMax	22.9	2.5	2.4	2.1	2.4	1.9	4.5	4.4	6.9	2.8	3.5	2.3	2.4
	PrecMin	14.4	0.6	0.3	0.3	0.2	0.6	0.4	0.7	0.7	0.6	0.8	0.4	0.5
	PrecStd	2.2	0.5	0.6	0.4	0.6	0.4	1.1	1.0	1.3	0.7	0.7	0.4	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.2	37.6	39.1	49.7	66.3	73.8	82.1	77.3	64.5	51.7	39.9	39.2
		MCWB	34.4	35.0	35.2	42.9	52.3	58.7	64.4	60.3	55.3	47.4	38.0	35.9
	2%	DB	33.8	34.6	36.7	43.7	59.0	67.4	75.4	71.2	60.4	48.6	37.4	36.0
		MCWB	31.8	32.5	33.7	38.3	48.1	55.2	60.5	59.2	53.1	45.6	35.9	33.9
	5%	DB	30.8	31.9	34.0	40.8	53.8	62.4	70.9	66.3	57.1	46.1	35.5	34.0
		MCWB	29.4	30.4	31.4	36.4	45.9	52.8	59.2	57.2	51.7	43.5	34.1	32.3
	10%	DB	28.4	29.0	31.9	38.5	49.4	57.6	66.1	62.3	53.9	43.5	33.9	32.0
		MCWB	27.3	27.7	29.8	35.0	43.2	49.6	57.4	55.5	49.7	41.3	32.7	30.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.8	35.4	35.6	42.6	54.2	60.7	65.6	63.8	56.8	48.1	38.3	36.3
		MCDB	36.5	37.2	38.1	48.4	63.5	69.0	76.4	73.1	62.5	50.8	39.5	38.6
	2%	WB	32.0	32.7	33.8	38.7	49.2	56.6	62.5	60.4	54.2	45.8	35.9	34.2
		MCDB	33.5	34.3	36.2	42.7	57.1	64.1	72.7	67.5	58.8	48.1	36.9	35.9
	5%	WB	29.5	30.3	31.5	36.8	46.6	53.6	59.8	58.0	52.1	43.6	34.2	32.5
		MCDB	30.7	31.9	33.6	39.9	53.3	61.9	68.5	64.8	56.2	45.6	35.1	33.9
	10%	WB	27.1	27.7	29.8	35.4	43.8	50.4	57.6	55.9	50.2	41.4	32.8	30.6
		MCDB	28.3	29.0	31.7	37.8	48.8	56.9	65.5	61.6	53.4	43.2	33.7	32.0
Mean Daily Temperature Range	5% DB	MDBR	11.7	11.4	10.8	10.2	10.3	10.9	12.2	11.2	9.4	6.7	8.9	10.5
		MCDBR	10.8	9.7	9.4	10.9	19.0	19.5	21.5	19.4	13.8	8.1	7.6	8.9
		MCWBR	10.1	9.0	8.0	7.4	10.8	9.7	9.7	9.1	8.0	6.5	7.1	8.3
	5% WB	MCDBR	10.7	9.8	9.1	9.7	17.7	18.6	20.0	16.8	12.2	7.5	7.2	8.8
		MCWBR	10.2	9.2	8.0	6.9	10.6	10.1	10.2	9.1	7.9	6.3	6.9	8.4
Clear-Sky Solar Irradiance	taub	0.204	0.250	0.298	0.326	0.349	0.345	0.360	0.344	0.309	0.265	N/A	N/A	
	taud	1.923	2.060	2.155	2.223	2.363	2.467	2.453	2.479	2.529	2.219	N/A	N/A	
	Ebn at Noon	5	149	222	255	264	270	260	251	230	145	0	0	
	Edh at Noon	1	13	24	31	31	29	29	25	18	11	0	0	
All-Sky Solar Radiation	RadAvg	3	105	513	1058	1407	1512	1385	947	505	147	0	0	
	RadStd	0	11	30	76	144	122	144	94	42	18	0	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+1.10	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page